Proposed multishift-reflection insertion

ProveIt integration preview

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1 Completed multishift reflection and resonant spectral jets

The report [1] continues the independent-shift centered-reflection and spectral-jet questions in the incoming continuous-depth Lerch work. Let $g(p) = \pi/\sin(\pi p)$, with its logarithm real on $(0,1)$. For $\operatorname{Re} R > 0$ and $\operatorname{Re} a_j > 0$, put

$$\mathcal{M}_R(\boldsymbol{s};\boldsymbol{a}) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} g(p)^R \prod_j (p+a_j-1)^{-s_j}\,dp, \quad \max(0, 1-\min_j \operatorname{Re} a_j) < c < 1.$$

It is entire in every order and holomorphic in every right-half-plane shift.

Completed reflection, real-shift display. For real $a_j > 0$, initially excluding $a_j = 1/2$, the centered integral must be supplemented by

$$\begin{aligned}\mathcal{M}_R = {} & \frac{1}{2\pi}\int_{\mathbb{R}} \left(\frac{\pi}{\cosh \pi t}\right)^R \prod_j (a_j - 1/2 + it)^{-s_j}\,dt \\ & + \frac{1}{\pi}\int_{1/2}^{\max_j(1-a_j)} g(x)^R \sin\left(\pi \sum_{x<1-a_j} s_j\right) \prod_j |x+a_j-1|^{-s_j}\,dx.\end{aligned}$$

The second integral is zero when its upper endpoint is at most $1/2$. Split at all branch points. For a coincident block B to the right of $1/2$, the initial ordinary domain is $\operatorname{Re}\sum_{j\in B} s_j < 1$. The complete sum then continues entirely in all orders and holomorphically through every collision. The full report gives the corresponding horizontal-cut formula for arbitrary complex shifts; cut heights can merge without an ordering ambiguity.

Mixed logarithmic overlap. Put $\ell_j = \log|x+a_j-1|$ and $\chi_j = \mathbf{1}_{x<1-a_j}$. At total zero order, every spectral jet is an ordinary logarithmic integral. The degree-three cut polynomial, divided by π, is

$$\chi_1\ell_2\ell_3 + \chi_2\ell_1\ell_3 + \chi_3\ell_1\ell_2 - \pi^2\chi_1\chi_2\chi_3.$$

The final overlap term cannot be omitted. At common shift and depth two, the completed degree-Q moment is $-Q\gamma_{Q-1}(a)$, for $Q \geq 1$. For two shifts $b = a+h$, $a > 0$, $|h| < a$, the completed bilogarithmic moment has the normally convergent expression

$$\Lambda_{2,(1,1)}(a,b) = -2\gamma_1(a) + \sum_{k\geq 1}(-h)^k\left(\zeta^{[1]}(k+1,a) + \frac{\zeta(k+1,a)}{k}\right).$$

The general mixed-jet formula uses unsigned Stirling coefficients and one anchor Stieltjes constant plus positive-integer Hurwitz-zeta jets.

Integer contacts and Hermite reflection. For positive integer orders n_j, $N = \sum_j n_j$, and $0 < \operatorname{Re} a_j < 1$,

$$\mathcal{M}_R(\boldsymbol{n};\boldsymbol{a}) - (-1)^N \mathcal{M}_R(\boldsymbol{n};\mathbf{1}-\boldsymbol{a}) = \sum_p \operatorname*{Res}_{z=p} \left(g(z)^R \prod_j (z + a_j - 1)^{-n_j} \right).$$

The residues are confluent Hermite divided differences. Centering an integer pole contributes half its residue, in addition to its symmetric Hadamard finite part. In particular $\mathcal{M}_R(1;1/2) = \pi^R/2$, not zero.

All centered spectral jets. The entire alternating family is the continuation of

$$Z_R(v) = \frac{1}{\Gamma(v)} \int_0^\infty t^{v-1} (2\cosh(t/2))^{-R}\, dt.$$

For $F_R(w) = \mathcal{M}_R(w;1/2)$, set $L = \log(2\pi)$ and $C^{\mathrm{o}}_{R,m} = (-1)^m \pi (2\pi)^{R+2m-1}/(2m)!$. The first odd spectral derivative is

$$F_R^{[1]}(2m+1) = C^{\mathrm{o}}_{R,m} \left((L + \gamma - H_{2m}) Z_R(-2m) - Z_R^{[1]}(-2m) \right).$$

Every higher odd or even derivative has an explicit finite Bell formula in the report. The necessary Z_R derivatives are retained; discrete odd-value formulas alone do not determine them.

Gamma–polylogarithm consequence. Writing $G_R(x) = \Gamma(R/2 + ix/(2\pi))\Gamma(R/2 - ix/(2\pi))$, one has the entire spectral identity

$$\begin{aligned} \int_{\mathbb{R}} G_R(x) \frac{\operatorname{Li}_s(ie^{-x/2}) - \operatorname{Li}_s(-ie^{-x/2})}{i}\, dx \\ = 8\Gamma(R)\pi^{s+1}\Gamma(1-s)\cos(\pi s/2) Z_{R+1}(1-s), \end{aligned}$$

with removable resonances. All fixed order and depth derivatives pass under the ordinary integral. At $R = 1$ the value simplifies to $4\pi s(2 - 2^{-s})\zeta(s+1)$, giving explicit Stieltjes-valued order jets.

Status. These are proved representation and branch-completion statements, not claims of arithmetic minimality or global historical priority. The classical Gamma transform, prior anchor expansion, and one-shift hyperbolic Stieltjes formula are credited inputs. The Gaussian S_6 and revised S_8 candidates remain unresolved. The supplied numerical comparisons are not interval certificates, and no proof-assistant formalization is asserted.

References

[1] *Contact-Complete Multishift Lerch Reflection*, research continuation prepared for Vladimir Reshetnikov, 10 October 2026. Full proofs and verification are in the accompanying article.